

Mohamed Hossam Hassan Nabil

Generative AI & Backend Engineer | Systems Architecture & LLM Integration

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PROFESSIONAL SUMMARY

Architecture-first Generative AI & Backend Engineer who designs systems before writing a single line of code. Experienced in translating complex requirements into clean, scalable architectures — spanning RAG pipelines, agentic multi-step workflows, voice AI systems, and LLM-integrated backends. Skilled at scoping and constraining LLMs (Claude/Anthropic API, OpenAI, and others) to defined Scopes of Work, ensuring model outputs stay aligned with system intent and production requirements. Deploys and optimizes LLM inference infrastructure on RunPod, and integrates ERP/CRM platforms (Odoo) with AI-driven backends. Brings a strong optimization mindset to every layer — from prompt engineering and retrieval strategies to API design and database architecture. Proficient in FastAPI and C#.NET backends, with hands-on experience in agentic orchestration, real-time database sync, and fine-tuned voice models. Collaborative, remote-ready, and driven by building systems that are not just functional — but structurally sound.

EXPERIENCE

Generative AI & Backend Engineer | Confidential

- **Architecture-first development:** Designed end-to-end system architectures for AI pipelines before implementation — defining data flows, component boundaries, retrieval strategies, and integration points across RAG, voice, and agentic systems.
- **Scoped and constrained LLMs** (Claude/Anthropic API, OpenAI, and others) to defined Scopes of Work — engineering system prompts, guardrails, and retrieval pipelines to keep model behavior aligned with product requirements and operational boundaries.
- Built and deployed **LLM inference infrastructure on RunPod**, optimizing GPU resource allocation, cold-start latency, and model serving configurations for production workloads.
- Designed and implemented **agentic AI workflows** — multi-step autonomous pipelines where LLMs reason, plan, and execute actions across APIs, databases, and external tools with minimal human intervention.
- Developed **RAG-based chatbots** (standard, corrective, and hybrid RAG) for dynamic question-answering, integrating LLMs with vector databases and live database APIs for real-time synchronized information retrieval.
- Fine-tuned custom **voice models using RVC** (Retrieval-based Voice Conversion) and TTS pipelines, enabling character-specific and persona-specific voice output for conversational AI systems.
- Built **STT and TTS pipelines** via FastAPI endpoints, powering fully voice-enabled AI interactions.
- Architected and optimized **backend workflows** using FastAPI and Django, connecting RAG architectures directly to live APIs and databases with real-time sync.
- Collaborated remotely following **Agile methodologies**, ensuring code quality, documentation, and scalable deployment.

Odoo ERP & CRM Developer

- Developed and customized **Odoo ERP and CRM modules** to streamline business operations, including sales pipelines, inventory management, and customer lifecycle tracking.
- Integrated **Odoo backends with external APIs and AI-driven services**, enabling automated data flows between ERP workflows and intelligent processing layers.
- Designed and optimized **database schemas and ORM models** within Odoo's framework, ensuring performant queries and maintainable module architecture.
- Built custom **automation rules and scheduled actions** within Odoo to reduce manual overhead and enforce business logic at the system level.

AI for Good Sandbox – Semi-Finalist | startAD | Abu Dhabi, UAE | 2025

- Selected as a semi-finalist across MENA for a **Google-supported AI for Good program** tackling 18 real-world challenges aligned with UN SDG 3.
- Assigned to Challenge #1 by the Abu Dhabi Early Childhood Authority: building an AI solution to empower UAE youth in life planning, family formation, and financial readiness.

- **Architected an LLM + RAG system** for personalized youth guidance — designing retrieval pipelines, knowledge scoping, and conversational flow before implementation began.
- Collaborated in a cross-functional, multi-national team over a four-week intensive program focused on rapid AI prototyping and social impact product design.

Generative AI Professional Internship | Digital Egypt Pioneers | July 2025

- Completed an intensive AI program covering **LLMs, GANs, Computer Vision, NLP, and Deep Learning** using tools including Hugging Face.
- Complemented technical training with Personal Branding and Career Planning modules.

AI and Backend Solutions | March 2025

- Designed system architectures for **AI-powered MVPs** leveraging LLMs and RAG pipelines for intelligent retrieval, dynamic content generation, and backend automation.
- **Scoped LLM behavior** to project-specific requirements, engineering prompts and retrieval chains to keep outputs within defined operational boundaries.
- Solutions addressed real-world challenges in data extraction, workflow optimization, and knowledge delivery.

NLP & Automation – ESG Report Summarizer | Jan 2024 – Feb 2024

- Built automated pipelines for **web scraping ESG reports**, extracting and classifying ESG content, and applying NLP-based summarization for actionable business insights.

Agentic AI Automation System – Odoo Sales Integration | Client Project

- **Architected and built a full agentic AI automation system** for a client, integrating an LLM-powered autonomous agent directly with **Odoo CRM and Sales modules** to automate end-to-end sales workflows.
- Designed the **multi-step agentic pipeline** from the ground up — mapping agent decision points, tool call sequences, fallback logic, and Odoo data flows before any code was written.
- Agent autonomously handled **lead qualification, CRM record creation/updates, sales stage progression, and follow-up scheduling** within Odoo based on real-time inputs and business rules.
- Integrated **LLM reasoning with Odoo's API layer**, enabling the agent to read pipeline state, make context-aware decisions, and write back structured updates without human intervention.
- Implemented **guardrails and SoW constraints** to ensure agent actions remained within defined operational boundaries, preventing out-of-scope mutations to CRM data.
- Delivered a production-ready system with logging, error handling, and human-in-the-loop escalation paths for edge cases outside the agent's confidence threshold.

PROJECTS

Agentic Workflow Orchestration

- Built **multi-step agentic AI pipelines** using Make (Integromat) and custom Python scripts, connecting LLMs, APIs, and data sources into fully autonomous systems. Designed orchestration logic — including branching, error handling, and state management — prior to any implementation.

Conversational AI & Chatbot Systems

- Developed and deployed chatbot systems leveraging **standard, corrective, and hybrid RAG architectures** for context-aware question-answering. Integrated fine-tuned TTS and RVC voice models for persona-specific voice output. Built domain-specific web scraping pipelines to populate vector databases for grounded retrieval.

Web Development

- Developed **Django and FastAPI applications** with authentication, database integration, and AI model inference — implementing clean route-based data flows and modular backend logic.

AI Graduation Project – Autonomous Vehicle | Valeo Collaboration

- Developed AI modules and CNNs for an **embedded self-driving car**; built spatial-environment response and auto-parking systems deployed on Raspberry Pi with modular, testable Python logic and real-time control integration.

TECHNICAL SKILLS

- **Generative AI & LLMs:** RAG (standard, corrective, hybrid), LLM prompt engineering, SoW scoping & guardrails (Claude/Anthropic API, OpenAI), embedding-based search, vector databases, agentic AI workflows, RVC voice fine-tuning, TTS/STT pipelines
- **LLM Infrastructure:** RunPod deployment & optimization, GPU resource management, model serving, inference latency tuning
- **Backend Development:** FastAPI, Django, RESTful API design, real-time database integration, live API synchronization, C#/.NET
- **ERP & CRM:** Odoo ERP & CRM — module development, workflow automation, ORM optimization, API integration
- **Frontend:** React
- **Python & Algorithms:** Clean architecture, debugging, dynamic programming, efficient algorithm design
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn
- **Databases:** SQL (PostgreSQL), NoSQL, query optimization, real-time sync for AI-driven applications
- **Tools & DevOps:** Git, GitHub, Hugging Face, Web Scraping, Make (Integromat), Raspberry Pi, Agile methodologies

CERTIFICATIONS & PUBLICATIONS

- **IEEE ICCA Publication:** Medical Deep Learning – Diabetic Retinopathy Classification. Processed 30,000+ images using Pandas/NumPy pipelines. Trained CNN models achieving 95% test accuracy.
<https://ieeexplore.ieee.org/document/10927999>
- Digital Egypt Pioneers – Generative AI Professional (Completed, 2025)
- startAD AI for Good Sandbox – Semi-Finalist Certificate, Abu Dhabi (2025)
- Data Science & AI Track Certification
- Embedded Systems Certification

Certificates available upon request.

EDUCATION

BSc – Electrical and Communications Engineering | The British University in Egypt

- Distinction with Honors | GPA: 3.7 / 4.0 | Ranked 5th in class | 2nd Best Graduation Project

LANGUAGES

Arabic – Native **English** – Fluent